

rect answer on a paper to marking a bubble sheet by colouring the correct bubble (optical mark reader on OMR concept). In OMR sheets, the answers are darkened either with HB pencil or black ball pen. The major difficulty with this kind of testing is the wear and tear of the answer sheet. These sheets may get tampered with in transit and the optical marker reader machine may give some incorrect results. To overcome such problems, an online quiz program can be implemented. In this article, we demonstrate how to design a simple quiz program using JavaScript. This resolves the issue of a paper-based test and the errors due to wear and tear of optical paper. It also facilitates instant results.

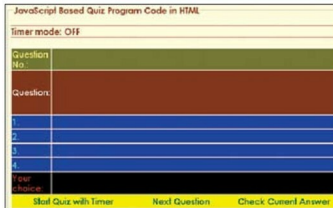
Software program

The program (Quiz.htm) is simple. It requires Internet Explorer as the browser. Since it is an objective-type quiz program, a developer has to define the questions as well as provide the correct answer in the source code, as given below.

```
function init()
{
  questions = new Array()
  questions[1] = "Which magazine does
  EFY Enterprises gets its name from?"
  questions[2] = "What significant
  event occurred in the space arena in
  1969?"
  questions[3] = "At present how many
  publications are brought out by EFY
  Enterprises?"
  questions[4] = "Present day powerful
  lighting devices?"
  questions[5] = "On Chandrayan-1,
  what does HySIC stand for?"

  answerA = new Array()
  answerA[1] = "Ethics For You"
  answerA[2] = "ESA was formed"
  answerA[3] = "6"
  answerA[4] = "Laser LEDs"
  answerA[5] = "Hyper Stereo Image
  Converter"

  answerB = new Array()
```



Screenshot 1: Program output of quiz program in JavaScript

```
answerB[1] = "Environment For You"
answerB[2] = "NASA was formed"
answerB[3] = "5"
answerB[4] = "Organic LEDs"
answerB[5] = "Hyper Sonic Integrated
Column"

answerC = new Array()
answerC[1] = "Electronics For You"
answerC[2] = "ISRO was formed"
answerC[3] = "4"
answerC[4] = "Mercury Vapour Lamp"
answerC[5] = "Hyper Spectral Imaging
Camera"

answerD = new Array()
answerD[1] = "Everything For You"
answerD[2] = "Man stepped on the
moon"
answerD[3] = "3"
answerD[4] = "Halogen Lamp"
answerD[5] = "Hyper Sonic Integrated
Camera"

rightAns = new Array
rightAns[1] = "3"
rightAns[2] = "4"
rightAns[3] = "1"
rightAns[4] = "1"
rightAns[5] = "3"
}

In this example, the ques-
tions are limited to just five. The
program can be extended to any
number of questions. If more ques-
tions are to be added, you have to
change the value '5' in the code
below.

var quizEnd = eval(5 * 1);
```

This program has four choices for the answer against each question, out of which only one is the correct answer. The program checks the input choice entered by the user and compares it with the correct answer

already stored in the code. The program output gives a letter grade, a percentage and the points for the quiz. If the choice of the answer tallies with the stored one, the counter increases the score and the question moves to the next one. Online help is also provided in the program.

This program has five 'Control Buttons', viz., 'Start Quiz with Timer', 'Next Question', 'Check Current Answer', 'Help', and 'Quit.' Each button has been assigned one or more functions, which help in the execution of the code. The sequence of clicking the buttons is as follows:

(For timer mode):

```
"Start Quiz with Timer" >> "Next
Question" >> "Choice" >> "Check
Current Answer" >> "Next Question".
```

(For normal mode):

```
"Next Question" >> "Choice" >>
"Check Current Answer" >> "Next
Question".
```

When the program is run, it will open up a window as shown in Screenshot 1. The program starts by clicking the 'Start with Timer' button. Here, for five questions, the time limit is set to 2 minutes. The timer value in seconds can be set in the program code:

```
secs =120 //Change this according to
your question duration
```

EFY note. The complete source code of this article has been included in this month's EFY-CD. ●